ASSIGNMENT – 01

DEPARTMENT OF SOFTWARE ENGINEERING

FACULTY OF COMPUTING

SABARAGAMUWA UNIVERSITY OF SRI LANKA

SE8106 – MOBILE COMPUTING

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Question 01

```
fun main() {
    print("Enter an integer: ")
    val number = readLine()!!.toInt()

    if (number > 0) {
        println("The number is Positive")
    } else if (number < 0) {
        println("The number is Negative")
    } else {
        println("The number is Zero")
    }
}
```

Question 02

```
fun main() {
    println("Numbers from 1 to 10:")
    for (i in 1..10) {
        println(i)
    }
    println("\nEven numbers from 1 to 20:")
    var num = 2
    while (num <= 20) {
        println(num)
        num += 2
    }
}
```

Question 03

```
fun addNumbers(a: Int, b: Int): Int {
    return a + b
}

fun main() {
    print("Enter first number: ")
    val a = readLine()!!.toInt()

    print("Enter second number: ")
    val b = readLine()!!.toInt()

    val result = addNumbers(a, b)
    println("Sum is: $result")
}
```

Question 04

```
fun main() {
    val numbers = arrayOf(10, 25, 5, 40, 15)
    var max = numbers[0]
    for (num in numbers) {
        if (num > max) {
            max = num
        }
    }

    println("Maximum value: $max")

    val numberList = numbers.toList()
```

```
    println("List: $numberList")
}
```

Question 05

```
class Student(val name: String, val age: Int, val marks: Int) {

    fun displayDetails() {
        println("Name: $name")
        println("Age: $age")
        println("Marks: $marks")
    }
}

fun main() {

    print("Enter student name: ")
    val name = readLine()!!

    print("Enter student age: ")
    val age = readLine()!!.toInt()

    print("Enter student marks: ")
    val marks = readLine()!!.toInt()

    val student1 = Student(name, age, marks)

    student1.displayDetails()
```

```
}
```

Question 06

```kotlin
open class Animal {

    open fun sound() {
        println("Animal makes a sound")
    }
}

class Dog : Animal() {

    override fun sound() {
        println("Dog barks")
    }
}

fun main() {

    val animal = Animal()
    animal.sound()

    val dog = Dog()
    dog.sound()
}
```

Question 07

```kotlin
package com.example.question07

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.material3.Button
import androidx.compose.material3.Text
import androidx.compose.runtime.*
import androidx.compose.foundation.layout.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContent {
            HelloApp()
        }
    }
}

@Composable
fun HelloApp() {
    var text by remember { mutableStateOf("Welcome") }

    Column(
        modifier = Modifier.fillMaxSize(),
        verticalArrangement = Arrangement.Center,
        horizontalAlignment = Alignment.CenterHorizontally
    ) {

        Text(text = text)

        Button(onClick = {
            text = "Hello Android"
        }) {
            Text("Click Me")
        }
    }
}
```

Question 08

```kotlin
package com.example.question08

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.text.input.TextFieldValue
import androidx.compose.ui.unit.dp

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContent {
            UserInputApp()
        }
    }
}

@Composable
fun UserInputApp() {

    var inputText by remember { mutableStateOf("") }
    var displayText by remember { mutableStateOf("") }

    Column(
        modifier = Modifier
            .fillMaxSize()
            .padding(16.dp),
        verticalArrangement = Arrangement.Center,
        horizontalAlignment = Alignment.CenterHorizontally
    ) {

        TextField(
            value = inputText,
            onValueChange = { inputText = it },
            label = { Text("Enter text") }
        )

        Spacer(modifier = Modifier.height(10.dp))

        Button(onClick = {
```

```
            displayText = inputText
        }) {
            Text("Submit")
        }

        Spacer(modifier = Modifier.height(10.dp))

        Text(text = displayText)
    }
}
```

Question 09